



- 1 (a) Find the values of k for which the system of equations

$$x + 2y + 3z = 1,$$

$$kx + 5y + 6z = 2,$$

$$7x + 2ky + 9z = 3,$$

does not have a unique solution.

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- (b) Given that $k = 1$, show that the system of equations in part (a) is consistent. Interpret this situation geometrically.

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